SafetySense
CALLS FOR HELP WHEN YOU CAN'T

ATTACHMENT 3 OF 5 · PURPOSE, AUDIENCE, FUNCTIONS, PATHWAY TO SATISFACTION

# A safety app that doesn't wait for you to press a button

During a night out, a walk home, a first date or a ride, SafetySense notices on the phone itself when something is wrong. If you can't respond, it brings in your trusted people and then professional dispatch. It supplements 911 and never replaces it.

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Status **working prototype**

## 01 Purpose

### Get help to a student when they can't get it themselves, without recording their life or asking them to give up their privacy.

The safety apps we reviewed (UNM Safe, Noonlight, bSafe and Apple's built-in Emergency SOS and Check In) mostly rely on the person pressing a button or running a timer. In a sexual assault that can fail:

**Freezing is normal**
In a study of 298 women seen after a sexual assault, 70% reported significant tonic immobility, an involuntary inability to move.[1]

**Drugging removes choice**
Someone who has been drugged may not recognise danger or be able to unlock a phone. No phone can detect a drug, but it can notice that you stopped responding.

**Phones get taken**
Most apps can't tell a dead phone from a quiet night. If SafetySense's heartbeat stops mid-session, its server raises the alert itself.

**No text-to-911 here**
New Mexico is still listed as the only US state without text-to-911. Its NG911 upgrade began in 2025.[2]

SafetySense keeps the basics students expect (SOS, live location, check-in timers, campus resources) and puts its effort into what's missing: **proactive, on-device detection** and a **calm, trauma-informed escalation ladder** that gets the right help without a tap.

## 02 Audience

| WHO | WHO THEY ARE | WHAT THEY NEED FROM SAFETYSENSE |
|---|---|---|
| **Primary users** | UNM and CNM students aged 18–25 who go out at night: parties, bars, concerts, first dates from dating apps, walks to a car or dorm, rideshares. | Protection that works even if they freeze or are drugged, privacy they can trust, and a setup that takes under 3 minutes. |
| **Trusted contacts** | Friends, roommates and family the user picks (1–5). They **don't need the app**. | A phone call and a text with a live link: where the person is, their battery, what happened, and one-tap "I'm on my way" or "Send help". |
| **Campus & community partners** | LoboRESPECT Advocacy Center, the UNM Women's Resource Center, UNMPD, CNM Security, and the Rape Crisis Center of Central NM. | A tool that routes students to advocacy, SANE care and reporting options on the student's terms. These are *proposed* partners and haven't been contacted yet. |

## 03 Functions

### Modes: the user turns guarding on for a specific situation

| MODE | WHEN | WHAT'S ON |
|---|---|---|
| **Everyday** | Always (default) | No microphone. Manual SOS, trusted contacts, medical info, campus resources. |
| **Walk Home** | Walking to a car, dorm or bus | Live location to a contact, an arrival check, and an alert if you leave the route or stop too long. |
| **Night Out** | Party, bar, concert | Full Guard: on-device sound and speech detection, motion, check-ins every 30–90 min, drink-check prompts. |
| **Date** | A first date from an app | Night Out plus **sealed date details** (name, profile, place), encrypted and shown to contacts *only if* an alert fires. |
| **Ride** | Uber, Lyft, someone else driving | Logs the plate and driver. Alerts on a route change or an unexpected stop longer than 3 minutes. |

### How it decides, all on the phone

- **Sound:** Apple's SoundAnalysis classifier listens for screaming, shouting, crying, gasps, slaps, thuds and breaking glass. Shouting counts for less when laughter or cheering is clearly present.
- **Words:** offline speech recognition catches the user's **safe word** (an instant alert) and distress phrases such as "stop", "get off me" and "let me go".
- **Motion and location:** a fall or impact followed by stillness, leaving where you said you'd be, or an off-route ride.
- **Responsiveness:** missed check-ins, and a server heartbeat that raises an alert if the phone goes dark.
- A **rules engine** scores these signals and can always explain why it escalated. On Apple Intelligence phones, an on-device language model gives a bounded second opinion. It can never be the only reason for an alert and can never block an SOS.

### The escalation ladder

| LEVEL | TRIGGER | WHAT HAPPENS | HOW TO STOP IT |
|---|---|---|---|
| 0 · GUARDING | Session on | Audio is held only in a 30-second memory buffer. Nothing is saved. Lock Screen Live Activity. | End session (PIN or Face ID) |
| 1 · CHECK-IN | Risk above threshold, or a scheduled check-in | A quiet haptic and "You okay?" with 45 seconds to answer. | Tap "I'm okay" |
| 2 · ALERT | No answer, high risk, safe word, Action Button, or lost heartbeat | Contacts get a call and a live link. Encrypted evidence capture starts. | Safety PIN |
| 3 · DISPATCH | 90 s at Alert with no cancel, a contact taps "Send help", or held SOS | Professional monitoring texts and calls the user, then, if there's no PIN or answer, passes the live location to 911. | Safety PIN, plus the dispatcher's call |

**Duress PIN:** it looks exactly like cancelling but silently goes to Level 3, for when someone forces the user to cancel. **Manual triggers:** Action Button, a Lock Screen and Control Center control, Siri ("SafetySense alert"), the safe word, and a hold-to-release SOS button. A **Fake call** helps the user leave a situation without explaining why.

### After an incident

The **Support** screen offers a 24/7 advocate, medical and SANE care, and reporting options (UNMPD, APD, Title IX or CNM), and also says plainly that keeping it private is fine. Nothing is required. The Face ID–locked **Vault** holds encrypted, hash-stamped evidence that the user shares only if they choose to.

### What no other app combines

| CAPABILITY | UNM SAFE | NOONLIGHT | BSAFE | APPLE BUILT-IN | SAFETYSENSE |
|---|---|---|---|---|---|
| Detects distress automatically | – | crash only (paid) | – | crash / fall | **sound, words, motion** |
| Alerts when the phone goes dark | – | – | – | Check In (may be delayed) | **server heartbeat** |
| Duress PIN | – | – | – | – | **yes** |
| Date details revealed only if needed | – | partial | – | – | **yes, encrypted** |
| Professional dispatch with live location | call to UNMPD | yes | – | 911 call | **yes (Noonlight API)** |
| UNM *and* CNM resources | UNM only | – | – | – | **both** |


SOURCES

1. Möller A., Söndergaard H. P., Helström L. "Tonic immobility during sexual assault: a common reaction predicting post-traumatic stress disorder and severe depression." *Acta Obstetricia et Gynecologica Scandinavica* 96(8):932–938, 2017. doi:10.1111/aogs.13174.
2. KFOX14, "New Mexico lacks technology allowing people to text 911 for an emergency"; Source NM, "New Mexico expected to launch new 911 system in August", Apr 11, 2025; GovTech, "Bernalillo County Leads New Mexico Next-Gen 911 Upgrade", Aug 11, 2025; Textdrip state-by-state guide, Jun 18, 2026 ("only one state ... doesn't support Text-to-911: New Mexico"). Checked Nov 4, 2026. Third-party relays such as accesSOS exist in Albuquerque and Santa Fe.
3. Comparison table: unmsafe.unm.edu; noonlight.com/noonlight-app; getbsafe.com/help; Apple iPhone User Guide (Check In, Crash Detection; Fall Detection is on Apple Watch). Checked Nov 4, 2026.

## 04 Pathway to satisfaction

We'll know SafetySense works when students trust it enough to turn it on, it rarely bothers them for nothing, and it gets help moving fast when it counts.

### Success metrics

| OUTCOME | TARGET | HOW WE MEASURE IT |
|---|---|---|
| **False alarms stay rare** | **< 1 per 10 h** | Alerts per 10 hours of Night Out guarding in pilot sessions, counted on the device and reported only with opt-in. |
| **Help moves fast** | **< 60 s** | Time from a scream in a scripted test to the first contact being called (event timestamps on the server). |
| **Battery is acceptable** | **< 8% / hour** | Measured with Xcode Energy tools on an iPhone 16e during 2-hour Guard sessions. |
| **Setup is quick** | **≥ 80% finish** | Share of pilot users who complete onboarding, including the practice drill, in under 3 minutes. |
| **Students feel safer, not watched** | **≥ 4 / 5** | Post-pilot survey: "I would turn this on for a night out" and "I trust what it does with my data." |
| **Contacts know what to do** | **≥ 90%** | Contacts in the practice drill who open the link and pick an action without instructions. |

### Validation and rollout

| | |
|---|---|
| DONE · NOV 2026 | **Working prototype.** A native iOS app with real on-device detection, a live backend, and real automated calls to a verified test phone. A dry-run test covers the full flow without calling anyone, and 136 automated tests cover the risk engine, escalation, vault and API contract. |
| DEC 2026 | **Detection lab tests.** Run the detector against recorded scenarios: parties, bars, arguments, screams and falls. Tune thresholds against the false-alarm and speed targets. |
| SPRING 2027 | **Students and advocates.** A survey and moderated usability tests with UNM and CNM students (onboarding and the practice drill). Ask LoboRESPECT and Rape Crisis Center of Central NM advocates to review every word of the alert and support copy. |
| FALL 2027 | **TestFlight pilot** with a student organization, with opt-in, privacy-preserving metrics only. Move from the Noonlight sandbox to production dispatch only after a legal and privacy review. |
| NEXT | **Campus partnership.** Offer it at new-student orientation and through the Women's Resource Center, with campus police numbers and SANE locations kept current by partners. |

FIT WITH THE CONTEST MISSION

SafetySense improves a real part of student life at UNM and CNM: getting home safely. It's built around local gaps, such as New Mexico's lack of text-to-911 and the move from LoboGuardian to UNM Safe, and around local resources. It also works for anyone, on any campus, who wants a friend in their pocket on a night out.